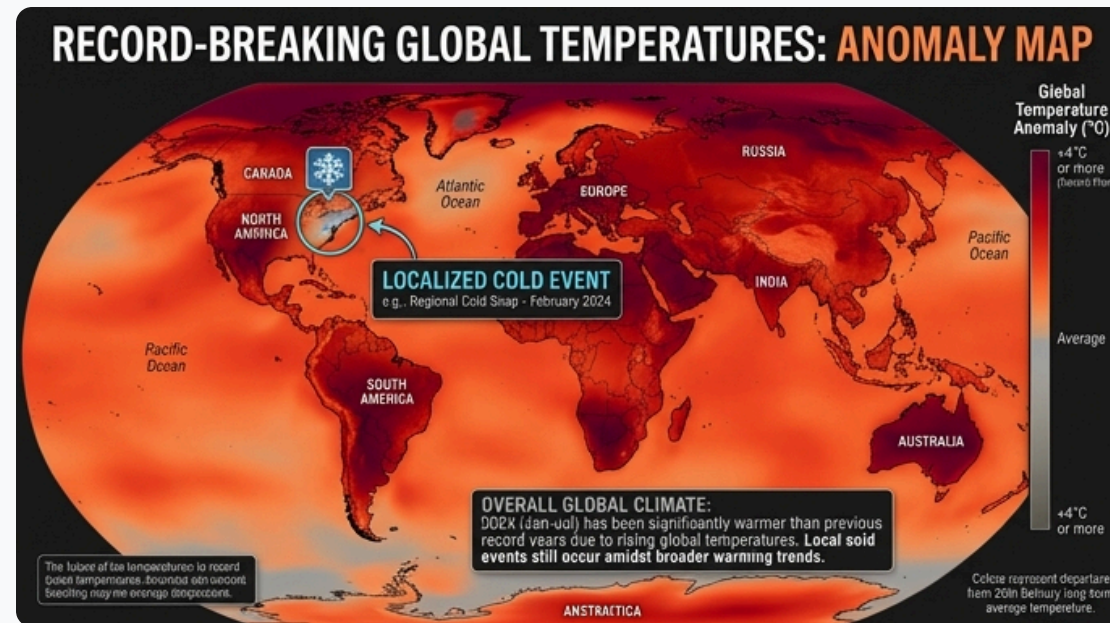


Is Global Warming Fake if it Snows?

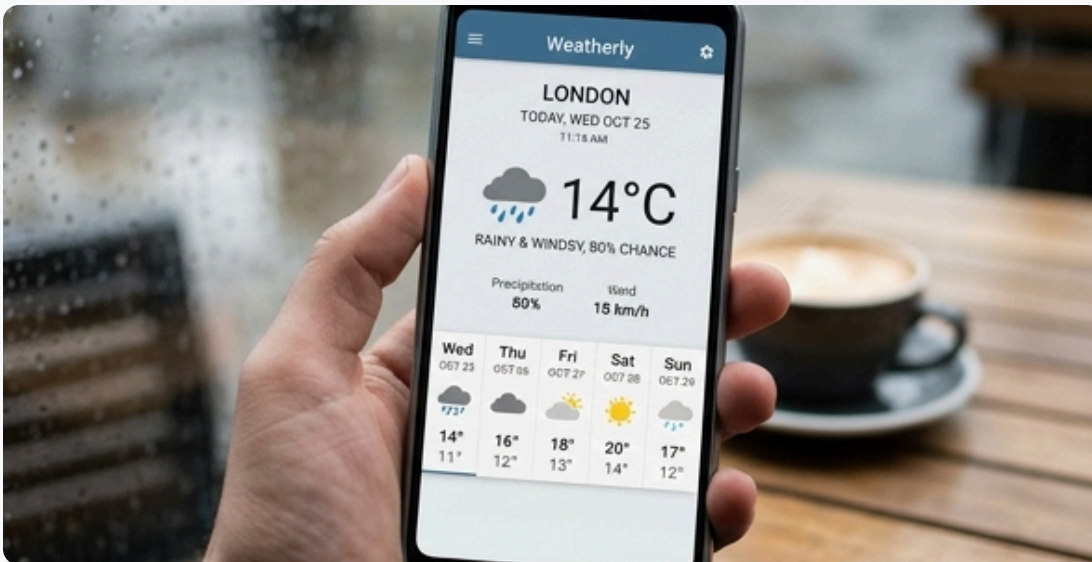
- Local snow doesn't negate global warming.
- A single event isn't a long-term trend.
- Proxy data shows 2023 was the hottest year.
- The baseline spans over 125,000 years.



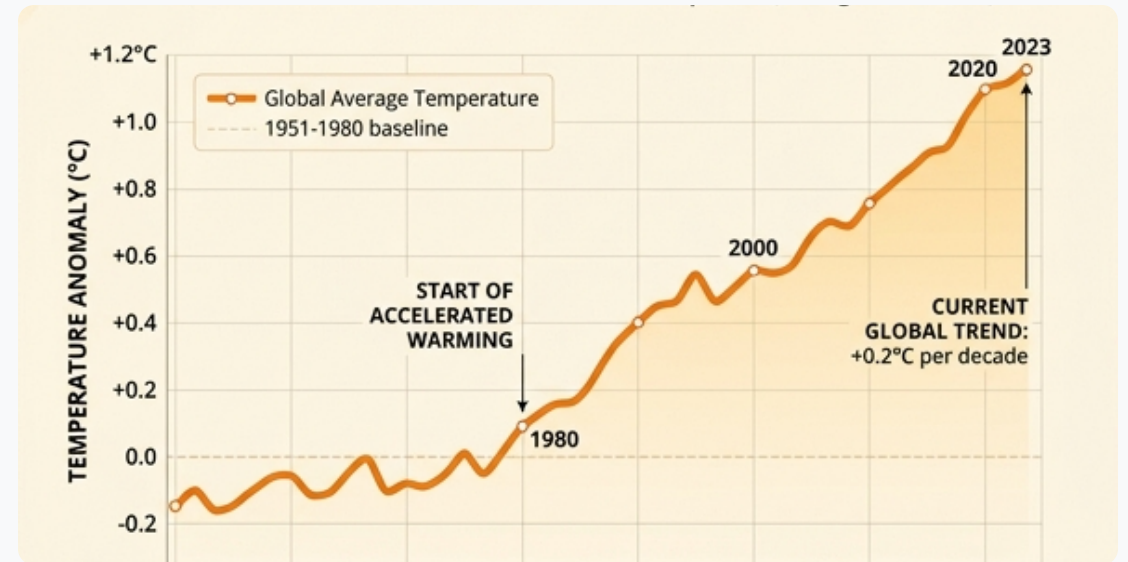
Global temperature anomalies show a long-term warming trend despite isolated cold weather events.

Defining Weather and Climate

- Weather = short-term, local conditions.
- Examples: Today's rain, tomorrow's wind.
- Climate = long-term statistical averages.
- Climate requires decades of data collection.



Weather describes the short-term, day-to-day atmospheric conditions in a specific location.



Climate represents the statistical average of weather data collected over thirty years or more.

Analogy: Mood vs. Personality

- Weather is your current mood.
- Moods change rapidly from day to day.
- Climate is your overall personality.
- Personality remains stable and predictable.



Your mood changes rapidly based on daily events, just like the weather.



Your personality is your stable, long-term baseline, just like the climate.

Evaluating Long-Term Trends

- Do not conflate single data points with trends.
- Climate relies on mathematical means: $\mu = \frac{1}{n} \sum_{i=1}^n x_i$
- Cold outliers don't erase a warming baseline.
- Always focus on decadal shifts, not daily events.

Identifying the Difference

- Distinguish between short-term events and long-term trends
- Avoid confusing localized conditions with global averages
- Use clear analogies to build a strong mental model



Weather represents the daily pages, while climate is the hourglass measuring decades.

Weather: Short-Term Events

- Sudden, highly localized atmospheric conditions
- Changes rapidly over minutes, days, or weeks
- Example: A sudden thunderstorm or a freezing weekend



A sudden thunderstorm is a classic example of weather: short-term and localized.

Climate: Long-Term Patterns

- Statistical average of weather over decades
- Establishes the baseline expectations of a region
- Example: The Sahara Desert averaging < 3 inches of rain annually



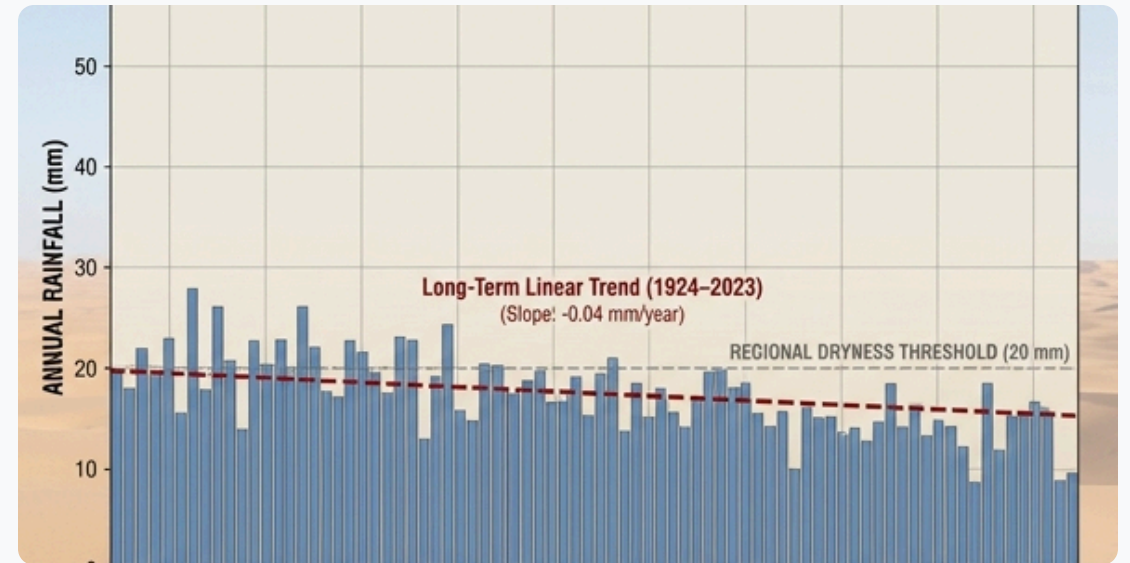
The Sahara's century-long dry streak is an example of climate, not just daily weather.

Weather vs. Climate in Action

- Weather = A freezing weekend anomaly in Texas
- Climate = A century of drought in the Sahara
- Non-example: A single hot day does NOT prove climate change



Weather: A localized, short-term temperature anomaly.



Climate: A century-long historical average establishing a trend.

The Cold Day Fallacy

- Does snow today mean global warming is fake?
- A very common climate misconception
- Confuses a local event with a global trend



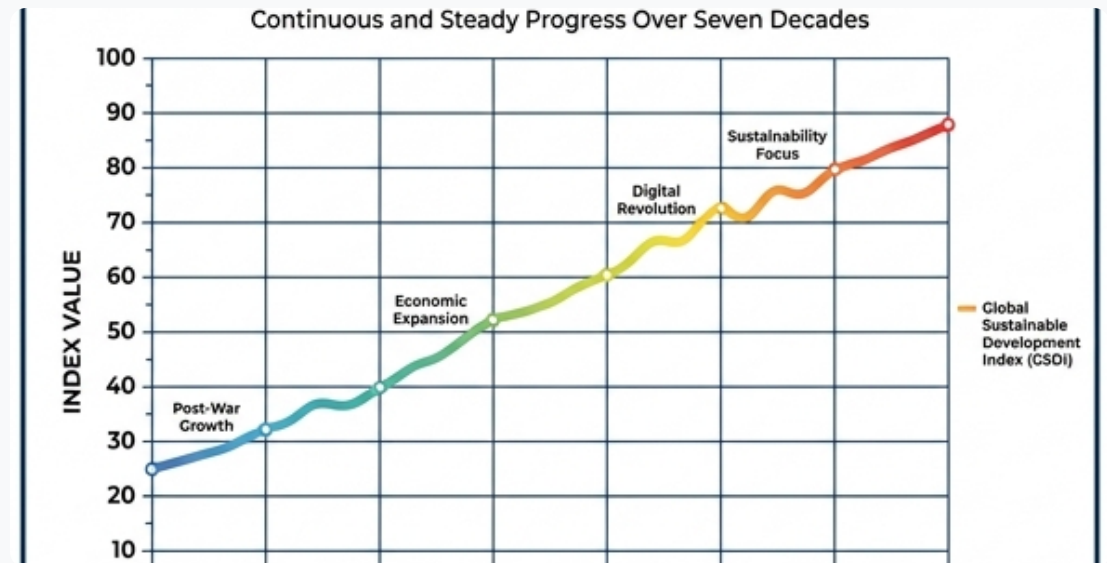
A local snowstorm is a short-term weather event, not an indicator of overall global climate trends.

Weather vs. Climate

- Weather: Short-term, local conditions
- Climate: Long-term statistical averages
- Analogy: Mood vs. Personality
- Weather changes daily; climate shifts over decades



Weather represents your temporary daily 'mood' and changes rapidly.



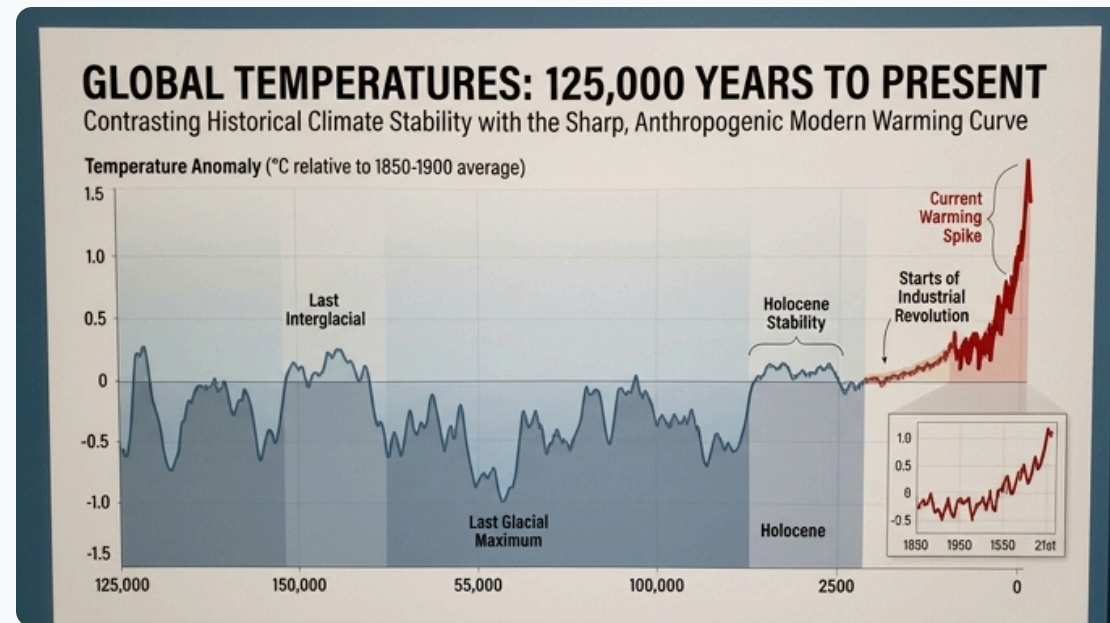
Climate represents your long-term 'personality' or historical baseline average.

The Snowstorm (A Non-Example)

- A single snowstorm is a strict non-example of climate
- It represents an isolated, local anomaly
- Cannot invalidate decades-long global warming trends
- Statistical averages outweigh single data points

The Baseline Reality

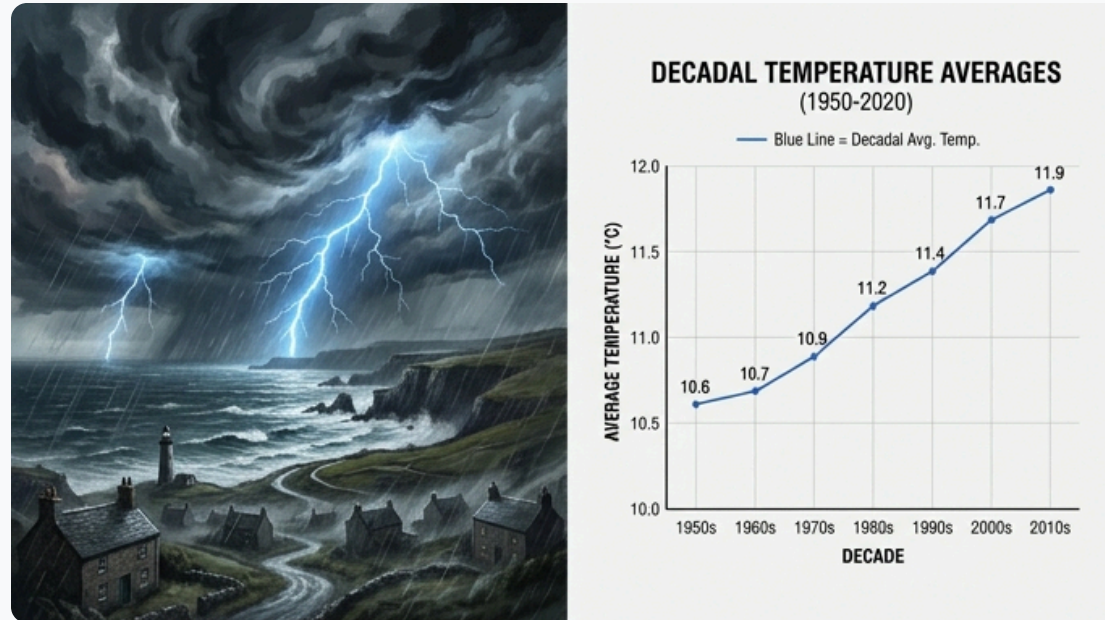
- Proxy data reveals long-term historical context
- Short-term weather obscures global reality
- 2023: The hottest year in over 125,000 years
- Global warming is a measured statistical trend



Historical proxy data demonstrates that 2023 broke global temperature records spanning millennia.

Classifying Meteorological Events

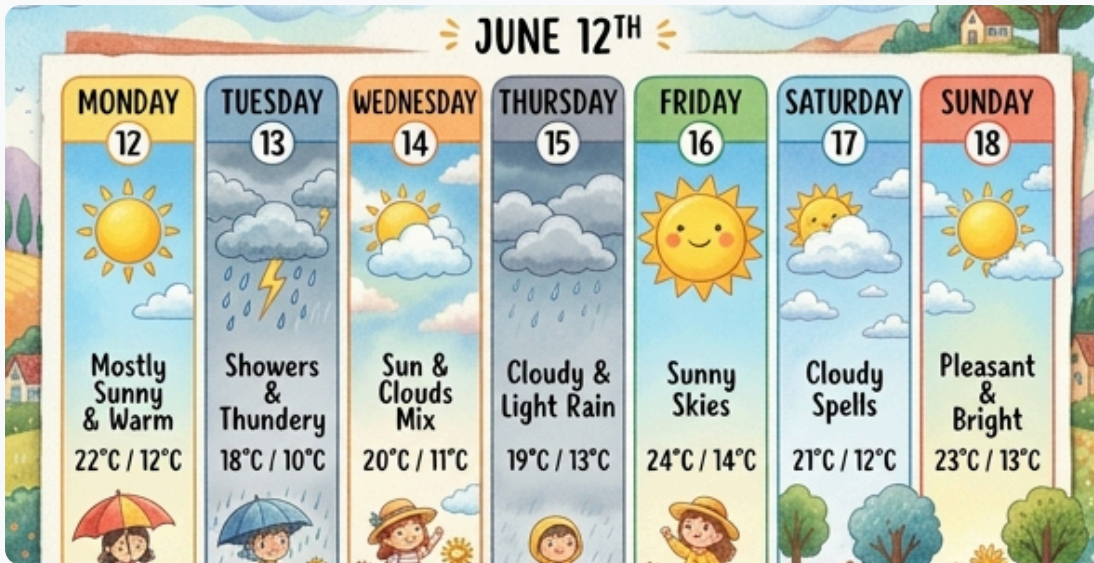
- Analyze real-world observations.
- Identify the underlying timeframe.
- Distinguish between isolated events and trends.



Weather represents immediate, chaotic conditions, while climate represents smooth, long-term statistical trends.

The Timeline Test

- Weather = Short-term conditions.
- Climate = Long-term statistical averages.
- Standard climate baseline is 30 years.



Weather tracks daily or weekly fluctuations in the atmosphere.



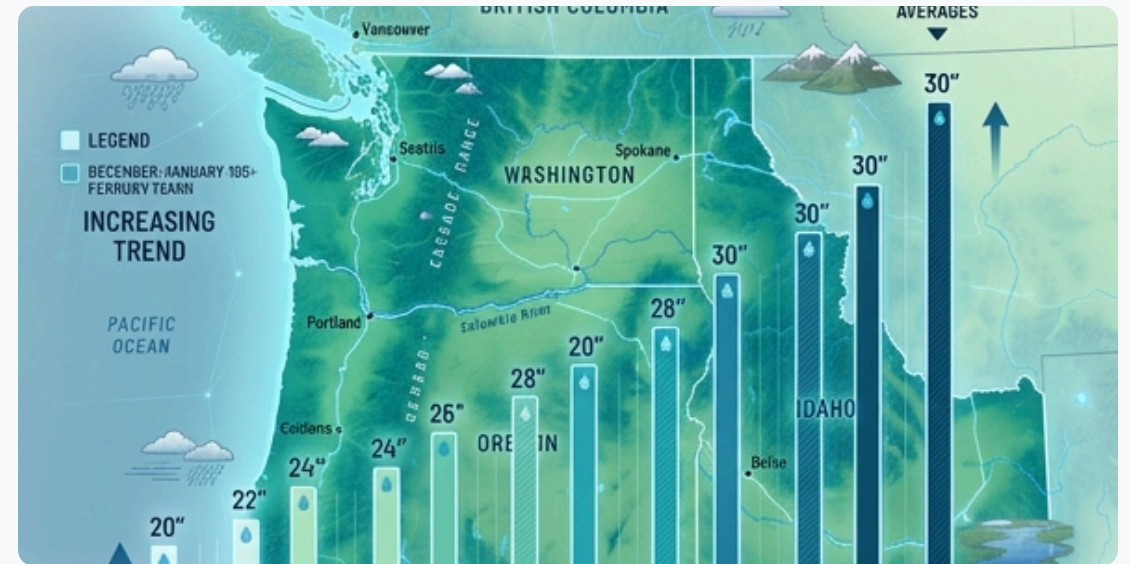
Climate tracks statistical averages over minimum 30-year periods.

Application: Analyzing a Trend

- Observation: Progressively wetter winters.
- Identify the timeframe: 30 years.
- Conclusion: This is a climate observation.



A single severe winter storm is classified as a weather event.



A 30-year trend of increasing precipitation is classified as a climate shift.

Non-Example: Misinterpreting Extremes

- Extreme events do not instantly redefine climate.
- A 100-year blizzard is still just weather.
- Isolated cold snaps do not disprove global warming.
- Always look for long-term statistical significance.